

3.9 Organic Synthesis

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.9 Organic Synthesis
Difficulty	Easy

Time allowed: 30

Score: /25

Percentage: /100

Question 1a

Alcohols and alkenes are versatile chemicals that are often involved in many organic synthetic reaction schemes.

Butan-1-ol can be converted directly into 1-bromobutane.

State the reagents and conditions required for this conversion.

[3 marks]

Question 1b

Heating butan-1-ol with concentrated sulfuric acid causes the loss of a small molecule.

State the type of reaction involved and name the resulting alkene.

[2 marks]

Question 1c

The alkene formed in part (b) can react with hydrogen bromide to form 1-bromobutane, however there will be a second organic product.

Name the second organic product formed.

[1 mark]

Question 2a

Describe one simple test-tube reaction that allows you to distinguish between compounds **G** and **H** shown in Fig. 2.1

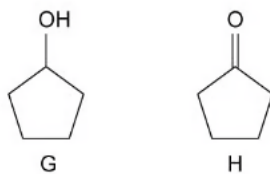


Fig. 2.1

Your answer should include a suitable reagent and the expected observations when each compound is tested separately with this reagent.

Reagent

Observation with **G**

Observation with **H**

[3 marks]

Question 2b

Describe one simple test-tube reaction that allows you to distinguish between compounds **I** and **J**, shown in Fig. 2.2.

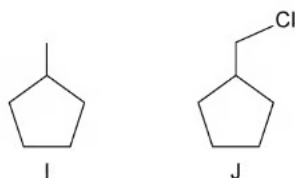


Fig. 2.2

Your answer should include a suitable reagent and the expected observations when each compound is tested separately with this reagent.

Reagents

Observation with **I**

Observation with **J**

[3 marks]

Question 2c

Describe one simple test-tube reaction that allows you to distinguish between compounds **K** and **L**, shown in Fig. 2.3.

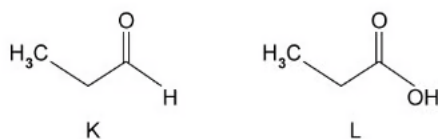


Fig. 2.3

The test-tube reaction should **not** convert compound **K** into compound **L**.

[3 marks]

Question 3a

A reaction pathway is shown in Fig. 3.1.

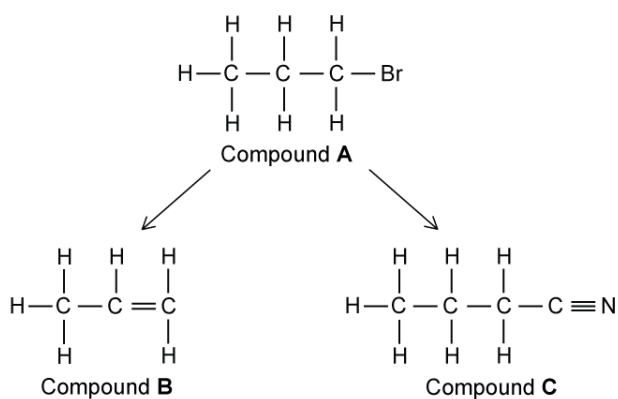


Fig. 3.1

State the reagents and conditions required for the conversion of compounds:

A to B

B to C

[6 marks]

[3 marks]

Model Answers: Easy

1a

a) The reagents and conditions required for the conversion of butan-1-ol into 1-bromobutane are:

- Sodium bromide / NaBr; [1 mark]
- Sulfuric acid / H₂SO₄; [1 mark]
- Heat under reflux; [1 mark]

OR

- Phosphorous tribromide / PBr₃; [1 mark]
- Dropwise addition / added one drop at a time; [1 mark]

[Total: 3 marks]

- Practice recalling the reagents, conditions, mechanisms and reaction types for all of these conversions as it will help with answering exam questions easily
 - It will also help with your A-level knowledge as there will be more conversions added

1b

b) The type of reaction involved when butan-1-ol is heated with concentrated sulfuric acid causing the loss of a small molecule is:

- Elimination

OR

Dehydration; [1 mark]

The resulting alkene is:

- But-1-ene; [1 mark]

[Total: 2 marks]

- In this case, the small molecule that is lost will be water
 - This means that the alkene formed will be but-1-ene
 - But-2-ene cannot be formed as the hydroxyl group is on carbon 1

1c

c) The second organic product formed when the alkene formed in part **(b)** reacts with hydrogen bromide is:

- 2-bromobutane; [1 mark]

[Total: 1 mark]

- Hydrogen bromide will undergo electrophilic addition, adding across the double bond of the alkene
- In this case, the alkene is but-1-ene
 - This means that the bromine can end up attached to carbon-1 or carbon-2
 - 1-bromobutane will be the minor product as the primary carbocation intermediate is less stable than the secondary carbocation intermediate that forms 2-bromobutane
- You could deduce that the alkene is but-1-ene as this would be the only feasible alkene able to produce 1-bromobutane

2a

a) A suitable reagent and expected observations are:

- Reagent = acidified potassium dichromate(VI) solution; [1 mark]
- Observation with **G** = colour change from orange to green; [1 mark]
- Observation with **H** = no visible change; [1 mark]

OR

- Reagent = (acidified) potassium manganate(VII) solution; [1 mark]
- Observation with **G** = colour change from purple to colourless; [1 mark]
- Observation with **H** = no visible change; [1 mark]

[Total: 3 marks]

- Compound **G** is a secondary alcohol which can be oxidised to a ketone
- Compound **G** will therefore change acidified potassium dichromate(VI) solution from orange to green
- Compound **H** is a ketone which will not react with acidified potassium dichromate(VI) solution
- The same applies to acidified potassium manganate(VII) solution but the colour change will be from purple to colourless

2b

b) A suitable reagent and expected observations are:

- Reagents = sodium hydroxide

AND

Followed by acidified silver nitrate solution; [1 mark]

- Observation with **I** = no visible change; [1 mark]
- Observation with **J** = forms a white precipitate; [1 mark]

[Total: 3 marks]

- The only difference between compounds **I** and **J** is the chlorine atom attached to the methyl group in compound **J**
- This can be tested using the halide ion test of acidified silver nitrate solution after the sample has been reacted with sodium hydroxide solution
 - A nucleophilic substitution reaction occurs with the sodium hydroxide which releases the chlorine as a chloride ion
- Chlorides give a white precipitate, which fully dissolves in dilute ammonia

2c

c) A suitable reagent and expected observations are:

- Reagent = sodium carbonate solution; [1 mark]
- Observation with **K** = no visible change; [1 mark]
- Observation with **L** = effervescence / bubbles of gas form; [1 mark]

[Total: 3 marks]

- Compound **K** is an aldehyde and compound **L** is a carboxylic acid

- **Careful:** It is very tempting to put down Fehling's or Tollens' as your answer but the question says that the reagent should **not** convert compound **K** into compound **L**
 - This means that you must test for the carboxylic acid, not the aldehyde
- This question is testing both your analysis and synthesis knowledge

3a

a) The reagents and conditions required are:

Conversion of **A** to **B**

- Ethanol; [1 mark]
- Potassium / sodium hydroxide; [1 mark]
- Heat under reflux; [1 mark]

Conversion of **A** to **C**

- Ethanol; [1 mark]
- Potassium cyanide / KCN; [1 mark]
- Heat under reflux; [1 mark]

[Total: 6 marks]

- The conversion of a halogenoalkane like compound **A** to an alkene requires an elimination reaction
- To form the nitrile from a halogenoalkane, a nucleophilic substitution reaction must occur

3b

b) The name of compound **C** is:

- Butanenitrile; [1 mark]

[Total: 1 mark]

- Potassium / sodium hydroxide; [1 mark]
- Heat under reflux; [1 mark]

Conversion of **A** to **C**

- Ethanol; [1 mark]
- Potassium cyanide / KCN; [1 mark]
- Heat under reflux; [1 mark]

[Total: 6 marks]

- The conversion of a halogenoalkane like compound **A** to an alkene requires an elimination reaction
- To form the nitrile from a halogenoalkane, a nucleophilic substitution reaction must occur

3b

b) The name of compound **C** is:

- Butanenitrile; [1 mark]

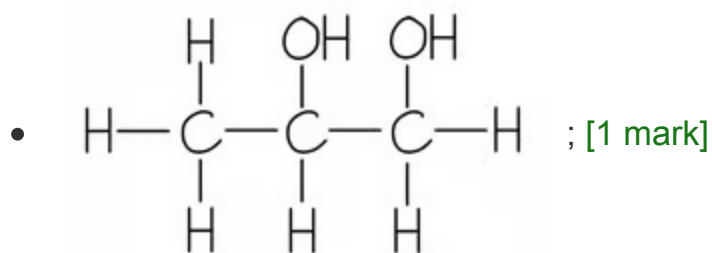
[Total: 1 mark]

- Nucleophilic substitution involving CN^- ions is an effective way of extending a carbon chain
- When naming a nitrile, ensure that:
 - Your spelling is correct and you do not miss the 'e' in butane
 - You count the correct number of carbons in the main carbon chain
 - Many students will incorrectly give the name of compound **C** as propanenitrile because the original organic molecule was propane-based

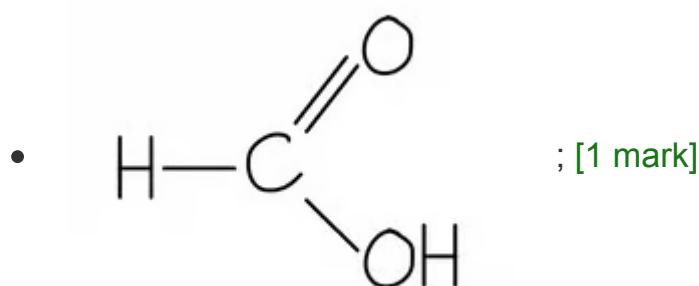
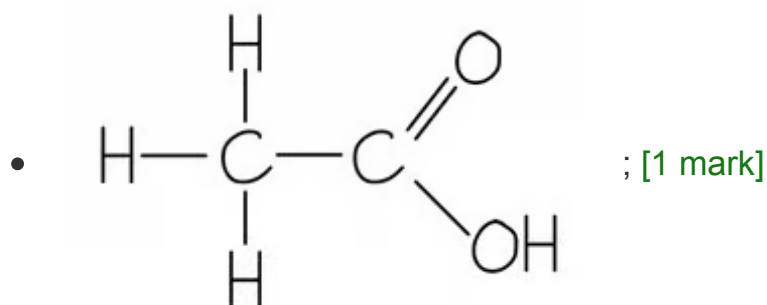
3c

c)

i) The structure of compound **D** is:

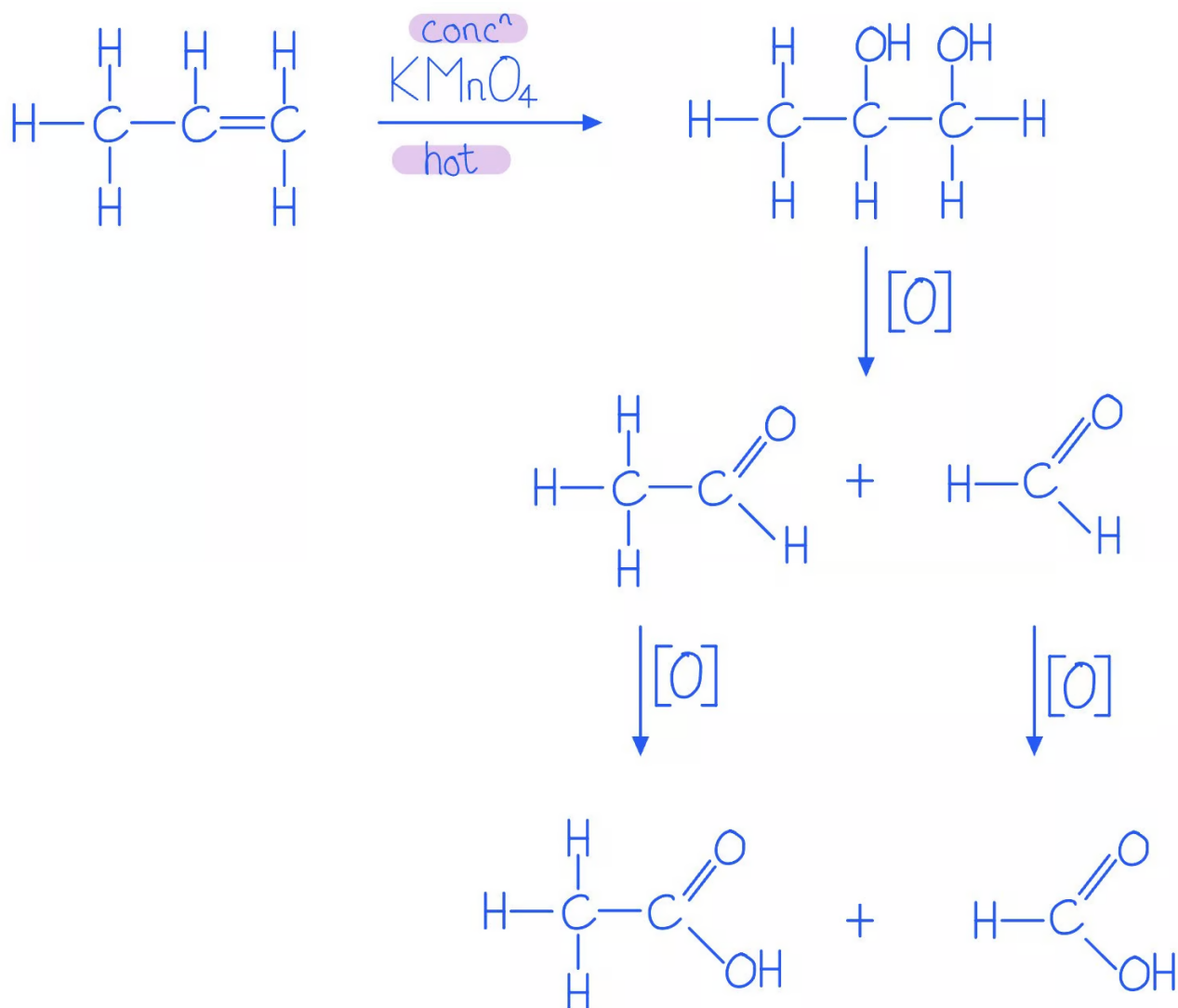


ii) The structure of compounds **E** and **F** are (in any order):



[Total: 3 marks]

- When an alkene reacts with potassium manganate(VII) the conditions will cause different products
- If **cold** and **dilute** a diol is formed
- If **hot** and **concentrated** the conditions are harsher causing the C-C double bond to completely break
- The O-H groups in the diol formed are further oxidised to ketones, aldehydes, carboxylic acids or carbon dioxide gas
- The actual products formed depend on what is bonded to the carbon atoms in the alkene



- In these conditions, the methanoic acid formed can be oxidised to carbon dioxide and water

